

✓ Alfido Tech — Internship Conversion Funnel Analysis

Business Question: *What are the key drivers of internship offer acceptance, and where in the recruitment funnel does Alfido Tech lose the most candidates?*

```
import os, pathlib
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display, Image
import warnings
warnings.filterwarnings('ignore')

plt.rcParams.update({'font.family': 'DejaVu Sans',
                    'axes.spines.top': False, 'axes.spines.right': False})

# — Auto-locate project root so paths work from any launch directory —————
def find_project_root():
    for p in [pathlib.Path.cwd()] + list(pathlib.Path.cwd().parents):
        if (p / 'data' / 'alfido_internship_funnel.csv').exists():
            return p
    return pathlib.Path.cwd()

PROJECT_ROOT = find_project_root()
os.chdir(PROJECT_ROOT)
DATA_PATH = str(PROJECT_ROOT / 'data' / 'alfido_internship_funnel.csv')
CHARTS_DIR = str(PROJECT_ROOT / 'charts')

print(f"Project root : {PROJECT_ROOT}")
print(f>Data file      : {DATA_PATH}")
print("Libraries loaded ✓")
```

```
Project root : /content
Data file    : /content/data/alfido_internship_funnel.csv
Libraries loaded ✓
```

✓ 1. Data Loading & Overview

```
df = pd.read_csv('alfido_internship_funnel.csv')
print(f"Dataset shape: {df.shape}")
print(f"Columns: {list(df.columns)}")
df.head(10)
```

Dataset shape: (500, 14)

Columns: ['applicant_id', 'application_month', 'department', 'source', 'education', 'cgpa', 'prior_experience_months', 'score']

	applicant_id	application_month	department	source	education	cgpa	prior_experience_months	score
0	ALC1000	Jun	Digital Marketing	LinkedIn	BCA	6.71		0
1	ALC1001	Dec	Web Development	College Portal	BCA	6.76		0
2	ALC1002	Sep	Web Development	Naukri	B.Tech	7.30		0
3	ALC1003	Aug	UI/UX Design	Company Website	B.Tech	7.83		0
4	ALC1004	Apr	Digital Marketing	Company Website	BCA	8.32		6
5	ALC1005	Apr	Data Science	Referral	BCA	6.78		0
6	ALC1006	Feb	Business Development	Referral	MBA	8.84		3
7	ALC1007	Oct	UI/UX Design	Company Website	MBA	7.26		6
8	ALC1008	Aug	Business Development	LinkedIn	MBA	7.48		0
9	ALC1009	Sep	Digital Marketing	College Portal	BSc	6.83		6

Next steps: [Generate code with df](#) [New interactive sheet](#)

2. Exploratory Data Analysis

```
print("=== Dataset Info ===")
df.info()
print("\n=== Missing Values ===")
print(df.isnull().sum())
print("\n=== CGPA Statistics ===")
print(df['cgpa'].describe())
```

```
=== Dataset Info ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   applicant_id           500 non-null    object
1   application_month       500 non-null    object
2   department             500 non-null    object
3   source                 500 non-null    object
4   education              500 non-null    object
5   cgpa                   500 non-null    float64
6   prior_experience_months 500 non-null    int64
7   screened               500 non-null    int64
8   interview_invited      500 non-null    int64
9   interview_attended     500 non-null    int64
10  task_assigned          500 non-null    int64
11  offer_made             500 non-null    int64
12  offer_accepted         500 non-null    int64
13  drop_reason            174 non-null    object
dtypes: float64(1), int64(7), object(6)
memory usage: 54.8+ KB
```

```
=== Missing Values ===
applicant_id      0
application_month 0
```

```

department      0
source          0
education       0
cgpa            0
prior_experience_months  0
screened        0
interview_invited  0
interview_attended  0
task_assigned   0
offer_made      0
offer_accepted  0
drop_reason     326
dtype: int64

```

```

=== CGPA Statistics ===
count    500.000000
mean      7.515520
std       0.878246
min       5.000000
25%       6.920000
50%       7.500000
75%       8.110000
max       10.000000
Name: cgpa, dtype: float64

```

```

for col in ['department', 'source', 'education']:
    print(f"\n=== {col.upper()} ===")
    print(df[col].value_counts())

```

```

=== DEPARTMENT ===
department
Data Science      165
Web Development   122
Digital Marketing  89
UI/UX Design      69
Business Development  55
Name: count, dtype: int64

```

```

=== SOURCE ===
source
LinkedIn          168
College Portal    106
Company Website   71
Referral          65
Naukri            59
Instagram         31
Name: count, dtype: int64

```

```

=== EDUCATION ===
education
B.Tech    186
BCA       75
MBA       69
BSc       68
MCA       53
BBA       49
Name: count, dtype: int64

```

3. Funnel Analysis

```

stages = ['applied', 'screened', 'interview_invited', 'interview_attended',
          'task_assigned', 'offer_made', 'offer_accepted']
counts = [len(df), df['screened'].sum(), df['interview_invited'].sum(),
          df['interview_attended'].sum(), df['task_assigned'].sum(),
          df['offer_made'].sum(), df['offer_accepted'].sum()]

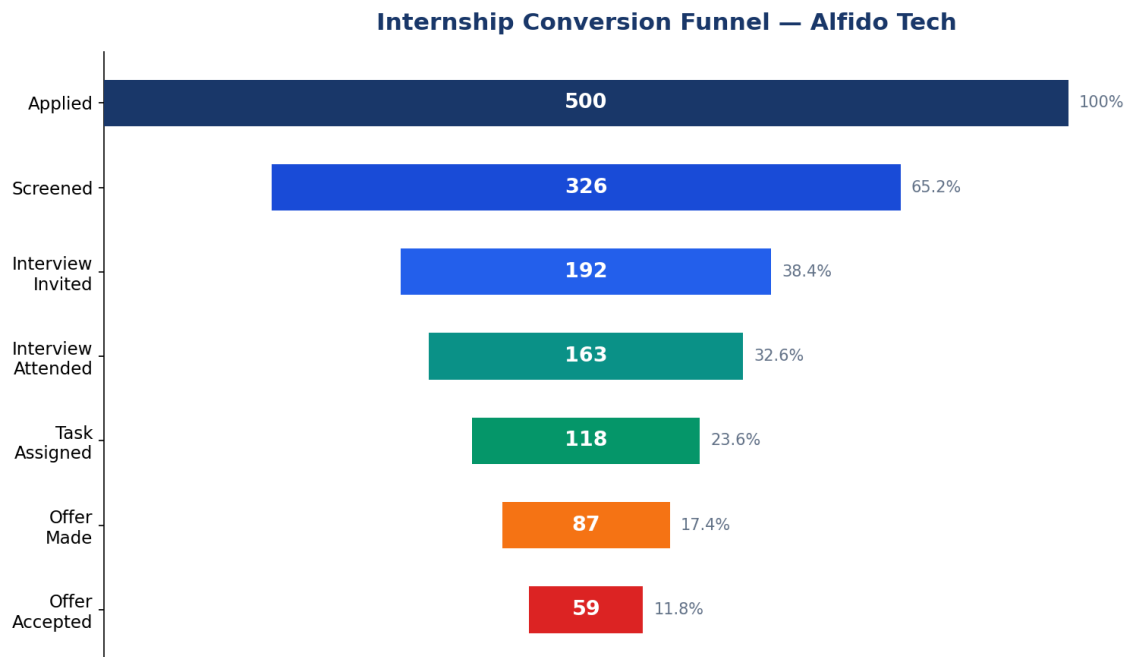
funnel_df = pd.DataFrame({
    'Stage': [s.replace('_', ' ').title() for s in stages],
    'Count': counts,
    'Overall Conv %': [round(c/counts[0]*100,1) for c in counts],

```

```
'Stage Conv %': [100.0] + [round(counts[i]/counts[i-1]*100,1) for i in range(1,len(counts))]
})
print(funnel_df.to_string(index=False))
```

	Stage	Count	Overall Conv %	Stage Conv %
	Applied	500	100.0	100.0
	Screened	326	65.2	65.2
	Interview Invited	192	38.4	58.9
	Interview Attended	163	32.6	84.9
	Task Assigned	118	23.6	72.4
	Offer Made	87	17.4	73.7
	Offer Accepted	59	11.8	67.8

```
Image(os.path.join(CHARTS_DIR, '01_funnel.png'))
```



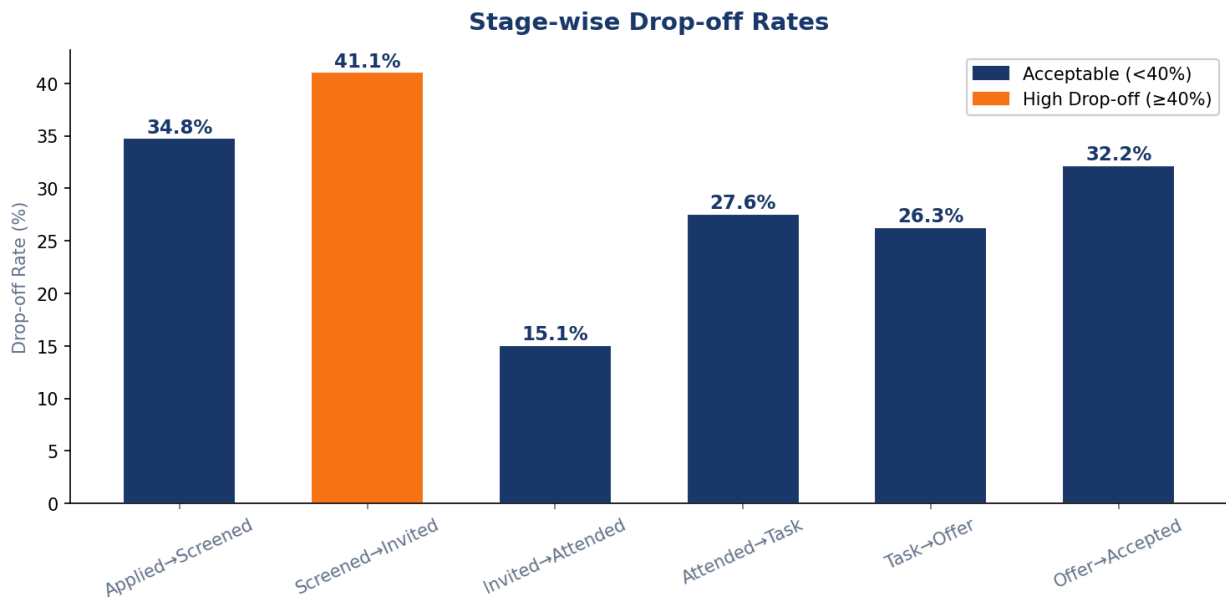
4. Stage-wise Drop-off Analysis

```
pairs = [('Applied','Screened',500,326),('Screened','Interview Invited',326,192),
        ('Interview Invited','Attended',192,163),('Attended','Task Assigned',163,118),
        ('Task Assigned','Offer Made',118,87),('Offer Made','Accepted',87,59)]
drop_df = pd.DataFrame([(a,b,n,m,round((n-m)/n*100,1)) for a,b,n,m in pairs],
                        columns=['From','To','Count In','Count Out','Drop-off %'])
print(drop_df.to_string(index=False))
print(f"\n⚠ Highest drop: Applied→Screened ({drop_df['Drop-off %'].max()}%)")
```

	From	To	Count In	Count Out	Drop-off %
	Applied	Screened	500	326	34.8
	Screened	Interview Invited	326	192	41.1
	Interview Invited	Attended	192	163	15.1
	Attended	Task Assigned	163	118	27.6
	Task Assigned	Offer Made	118	87	26.3
	Offer Made	Accepted	87	59	32.2

⚠ Highest drop: Applied→Screened (41.1%)

```
Image(os.path.join(CHARTS_DIR, '06_dropoff.png'))
```



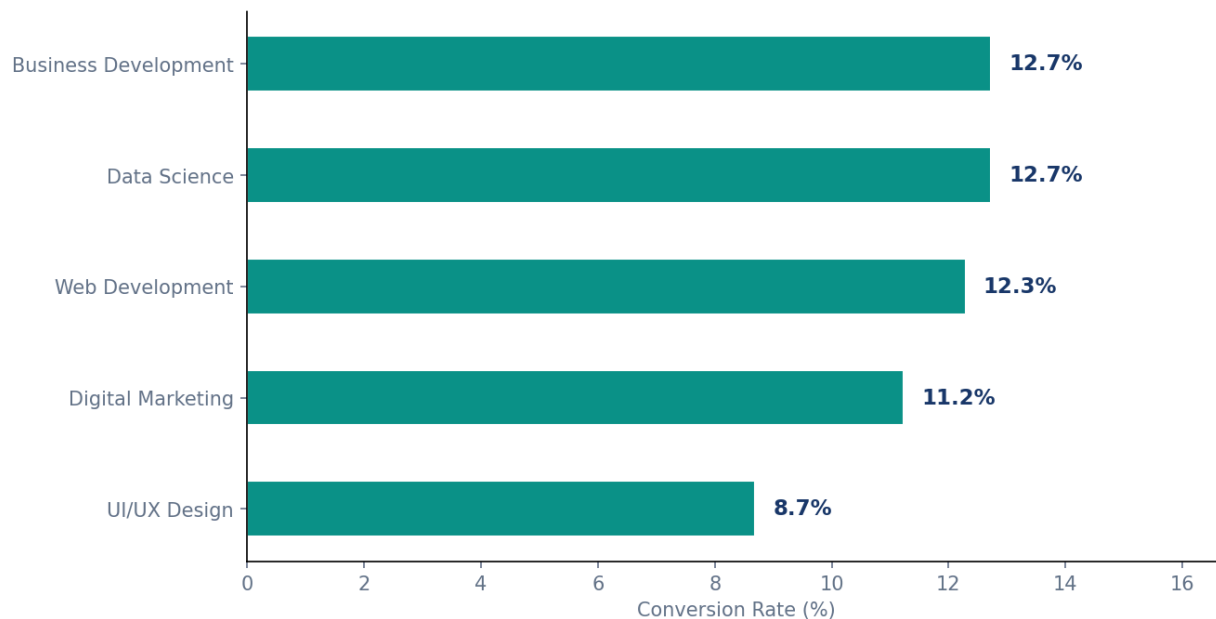
5. Department-wise Conversion

```
dept_stats = df.groupby('department').agg(  
    Applied=('applicant_id','count'),  
    Screened=('screened','sum'),  
    Offers_Made=('offer_made','sum'),  
    Accepted=('offer_accepted','sum')  
)  
.reset_index()  
dept_stats['Screen_Rate%'] = (dept_stats['Screened']/dept_stats['Applied']*100).round(1)  
dept_stats['Conv_Rate%'] = (dept_stats['Accepted']/dept_stats['Applied']*100).round(1)  
dept_stats = dept_stats.sort_values('Conv_Rate%', ascending=False)  
print(dept_stats.to_string(index=False))
```

department	Applied	Screened	Offers_Made	Accepted	Screen_Rate%	Conv_Rate%
Business Development	55	40	12	7	72.7	12.7
Data Science	165	111	27	21	67.3	12.7
Web Development	122	73	19	15	59.8	12.3
Digital Marketing	89	60	18	10	67.4	11.2
UI/UX Design	69	42	11	6	60.9	8.7

```
Image(os.path.join(CHARTS_DIR, '02_dept_conversion.png'))
```

Offer Acceptance Rate by Department



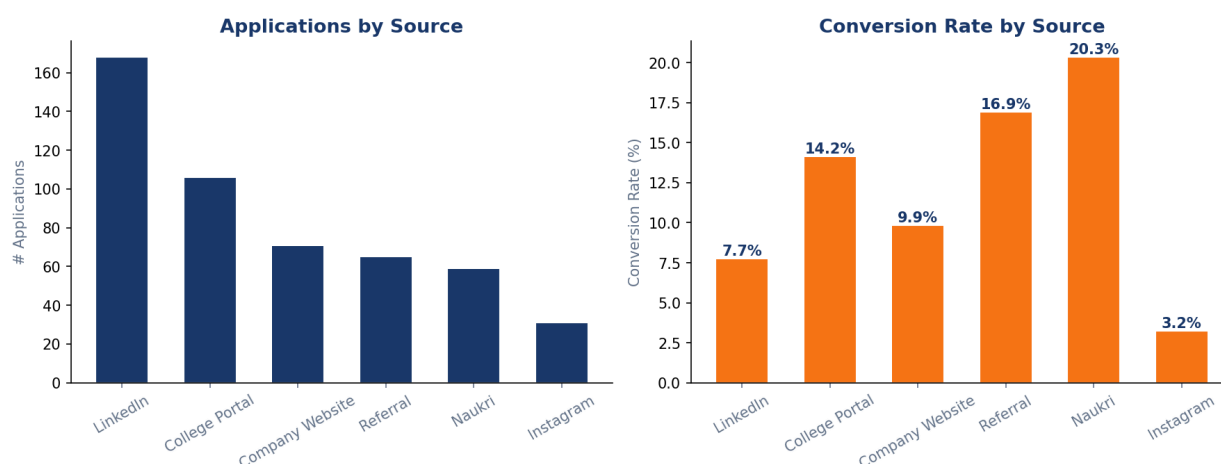
6. Recruitment Source Effectiveness

```
src_stats = df.groupby('source').agg(  
    Applied=('applicant_id', 'count'),  
    Accepted=('offer_accepted', 'sum')  
) .reset_index()  
src_stats['Conv_Rate%'] = (src_stats['Accepted']/src_stats['Applied']*100).round(1)  
src_stats['Efficiency'] = src_stats['Conv_Rate%'].apply(  
    lambda x: 'High' if x > 14 else ('Medium' if x > 10 else 'Low'))  
print(src_stats.sort_values('Conv_Rate%', ascending=False).to_string(index=False))
```

source	Applied	Accepted	Conv_Rate%	Efficiency
Naukri	59	12	20.3	High
Referral	65	11	16.9	High
College Portal	106	15	14.2	High
Company Website	71	7	9.9	Low
LinkedIn	168	13	7.7	Low
Instagram	31	1	3.2	Low

```
Image(os.path.join(CHARTS_DIR, '03_source.png'))
```

Recruitment Source Analysis

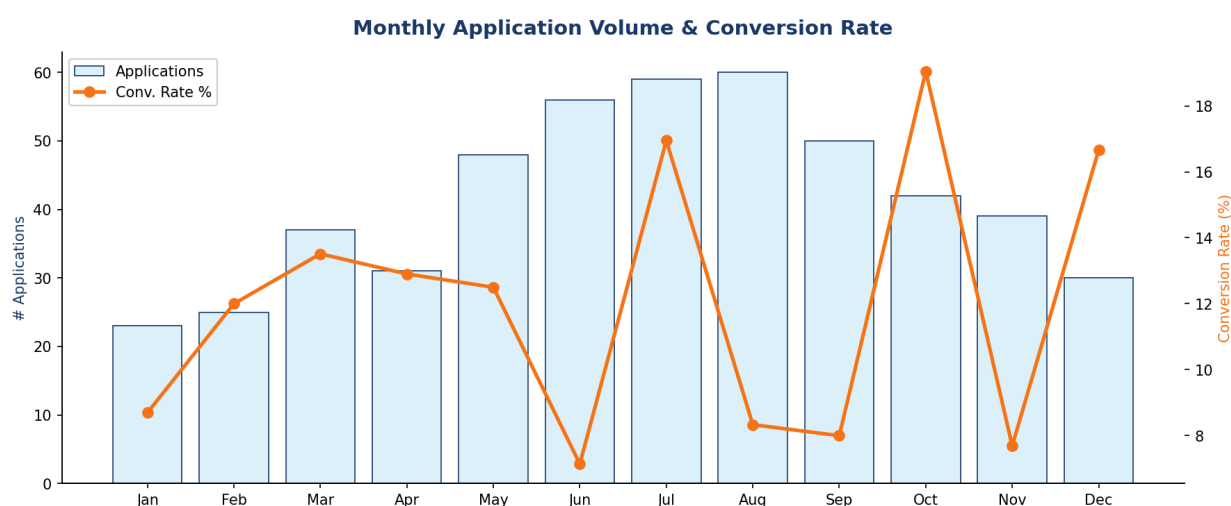


7. Monthly Trend Analysis

```
month_order = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
monthly = df.groupby('application_month').agg(
    Applications=('applicant_id', 'count'),
    Accepted=('offer_accepted', 'sum')
).reindex(month_order).fillna(0).reset_index()
monthly.columns = ['Month', 'Applications', 'Accepted']
monthly['Conv_Rate%'] = (monthly['Accepted']/monthly['Applications']*100).round(1)
print(monthly.to_string(index=False))
```

Month	Applications	Accepted	Conv_Rate%
Jan	23	2	8.7
Feb	25	3	12.0
Mar	37	5	13.5
Apr	31	4	12.9
May	48	6	12.5
Jun	56	4	7.1
Jul	59	10	16.9
Aug	60	5	8.3
Sep	50	4	8.0
Oct	42	8	19.0
Nov	39	3	7.7
Dec	30	5	16.7

Image(os.path.join(CHARTS_DIR, '04_monthly_trend.png'))

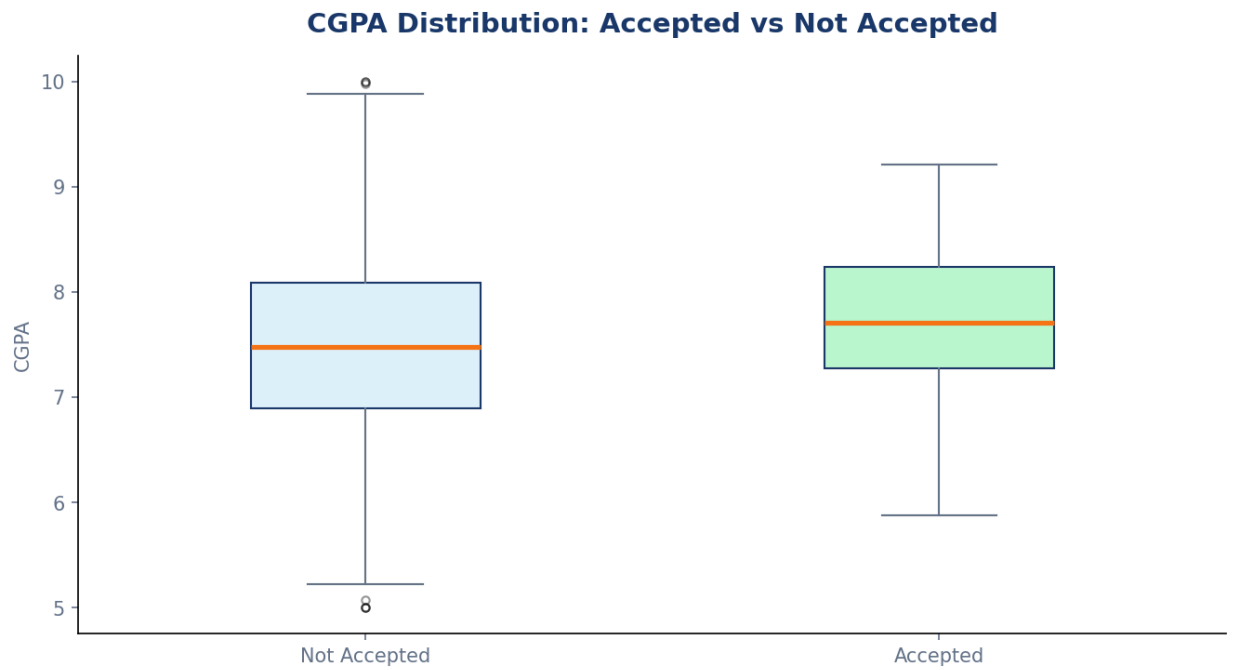


8. Academic Performance & Conversion

```
df['cgpa_band'] = pd.cut(df['cgpa'], bins=[5,6,7,8,9,10],
    labels=['5-6', '6-7', '7-8', '8-9', '9-10'])
band_stats = df.groupby('cgpa_band', observed=True).agg(
    Count=('applicant_id', 'count'),
    Accepted=('offer_accepted', 'sum')
).reset_index()
band_stats['Conv_Rate%'] = (band_stats['Accepted']/band_stats['Count']*100).round(1)
print(band_stats.to_string(index=False))
```

cgpa_band	Count	Accepted	Conv_Rate%
5-6	21	2	9.5
6-7	111	5	4.5
7-8	226	33	14.6
8-9	118	16	13.6
9-10	21	3	14.3

```
Image(os.path.join(CHARTS_DIR, '05_cgpa_boxplot.png'))
```



9. Key Insights & Recommendations

 Funnel Summary